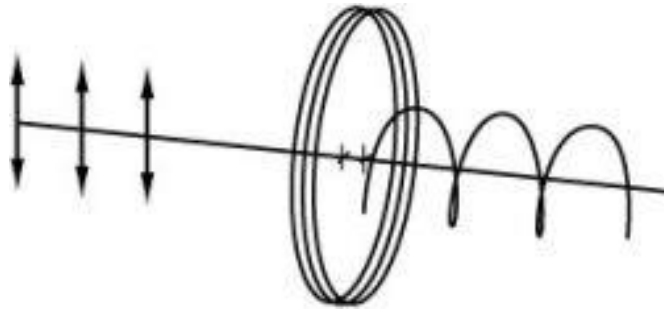


7.7 Retarders

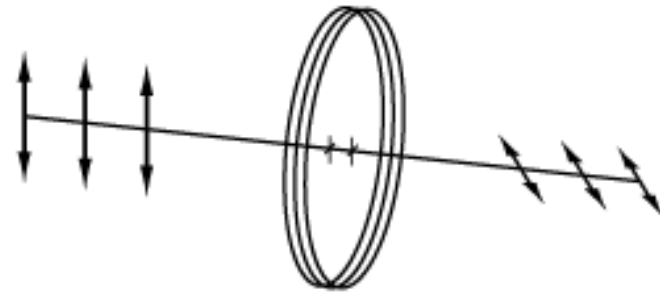
Retarders: the devices changing the polarization of an incident wave.

Principle: one of the two constituent coherent $\mathcal{P}$ -states is caused to have its phase lag behind that of the other by a predetermined amount.

Example:



Quarter-Wave Plates Change Linearly Polarized Light into Circularly Polarized Light.



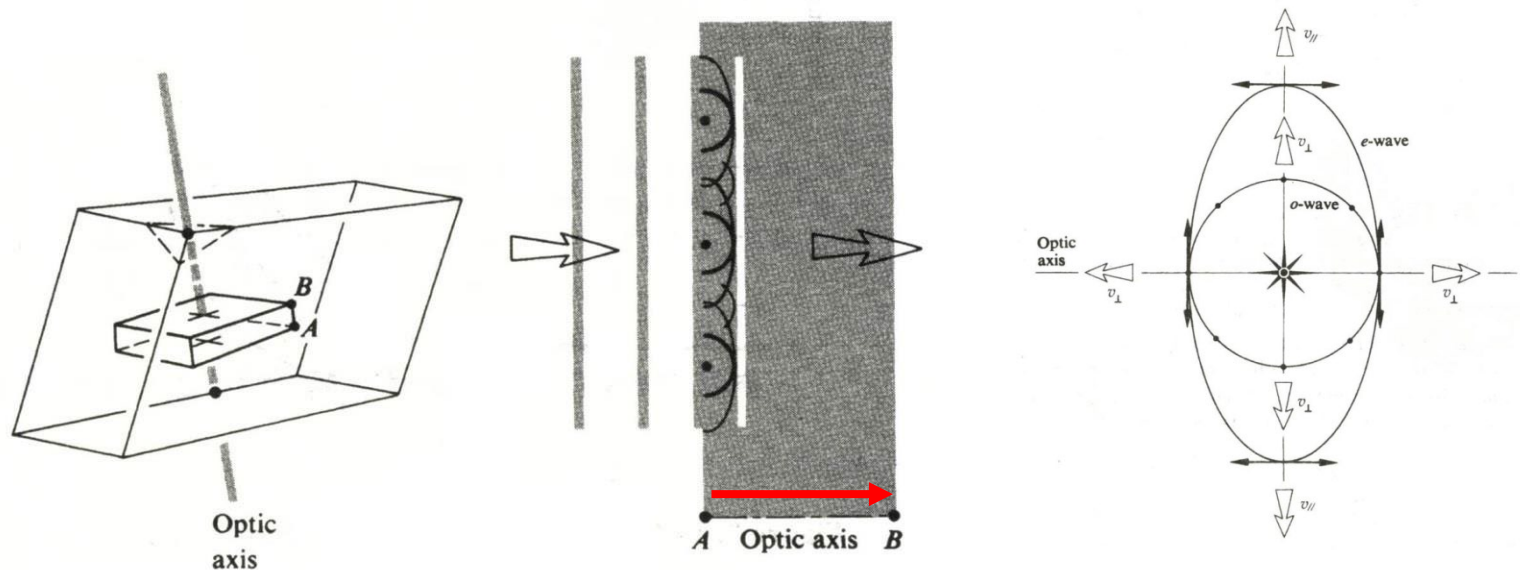
Half-Wave Plates Rotate the Polarization Plane of Linearly Polarized Light.

7.7.1 Wave Plates and Rhombs

I) The optic axis of calcite crystal is normal to both the front and back surfaces.

A normally incident plane wave:

- ❑ Only E-field perpendicular to the optic axis exists;
- ❑ The secondary spherical and ellipsoidal wavelets will be tangent to each other in the direction of the optic axis;
- ❑ The o- and e-waves will be coincident and a single undeflected plane wave will pass through the crystal. **No relative phase shift and no double images.**



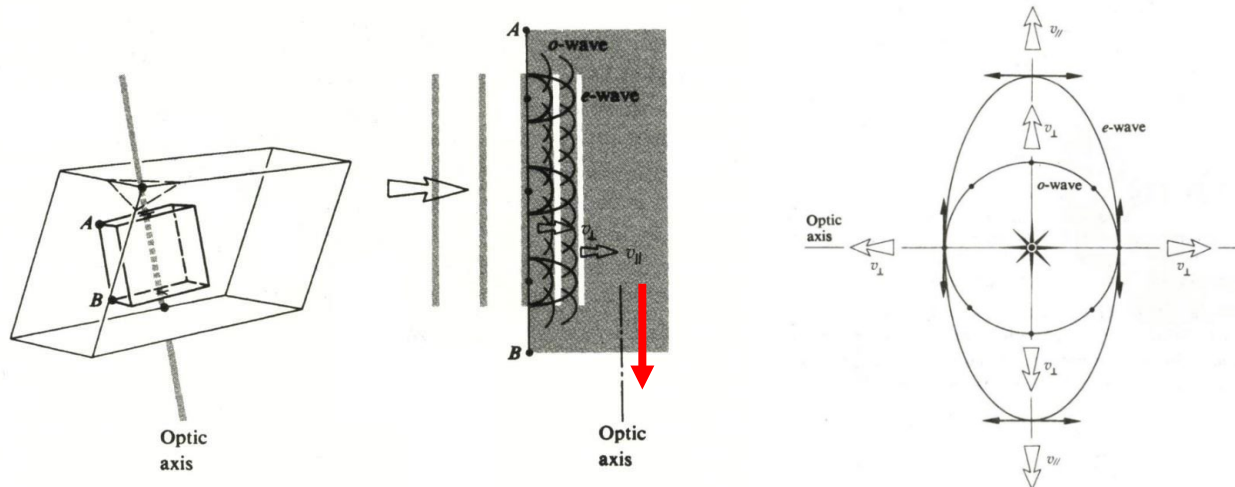
II) The direction of optic axis is parallel to the front and back surfaces.

- ❑ The E-field of an incident monochromatic plane wave has components parallel and perpendicular to the optic axis;
- ❑ Since $v_{\parallel} > v_{\perp}$, $n_o > n_e$, the e-wave will move across the specimen more rapidly than the o-wave.
- ❑ The relative optical path length [$\Lambda = d|n_o - n_e|$] and the phase difference [$\Delta\varphi = \frac{2\pi}{\lambda_0} d(|n_o - n_e|)$] are different.

$$\vec{E}_o(z, t) = \hat{i}E_{0o} \cos(kz - \omega t)$$

$$\vec{E}_e(z, t) = \hat{j}E_{0e} \cos(kz - \omega t + \Delta\varphi)$$

The state of polarization of the emergent light evidently depends on the amplitudes of the incoming orthogonal field components and of course on $\Delta\varphi$



The Full-Wave Plate:

The phase difference: $\Delta\varphi = \frac{2\pi}{\lambda_0} d(|n_o - n_e|)$

The Full-Wave Plate: $\Delta\varphi = 2\pi$, the relative retardation is one wavelength.

Note:

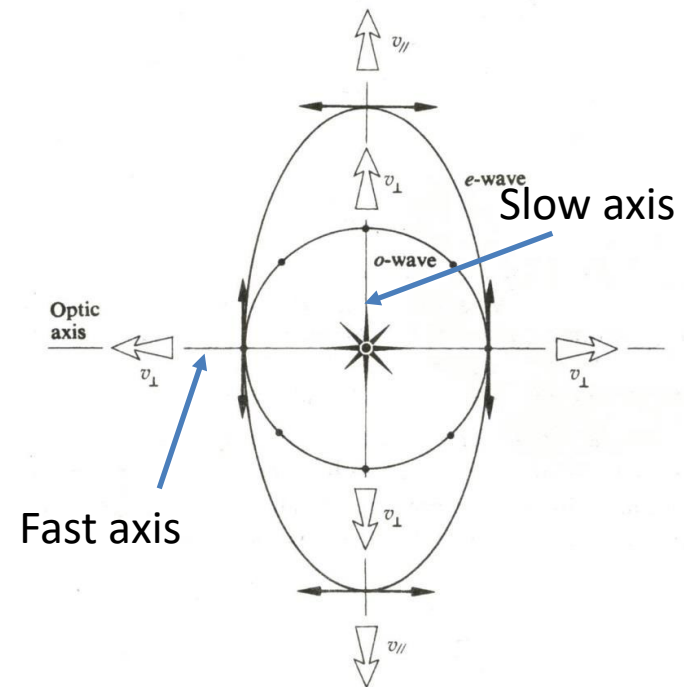
- $|n_o - n_e|$ changes little over the optical range;
- $\Delta\varphi \propto 1/\lambda_0$: a full-wave plate can function only in the manner discussed for a particular wavelength.

Calcite $v_{\parallel} > v_{\perp}$, $n_e < n_o$, $\Delta n = -0.172$, negative uniaxial.

Fast axis: the direction of optic axis in a negative uniaxial retarder is referred to as the fast axis.

Slow axis: the direction perpendicular to optic axis.

For positive uniaxial crystals: slow axis corresponds to the optic axis.



The Half-Wave Plate

The phase difference: $\Delta\varphi = \frac{2\pi}{\lambda_0} d(|n_o - n_e|)$

The Half-Wave Plate: $\Delta\varphi = \pi$, the relative retardation is $1/2$ wavelength.

Note:

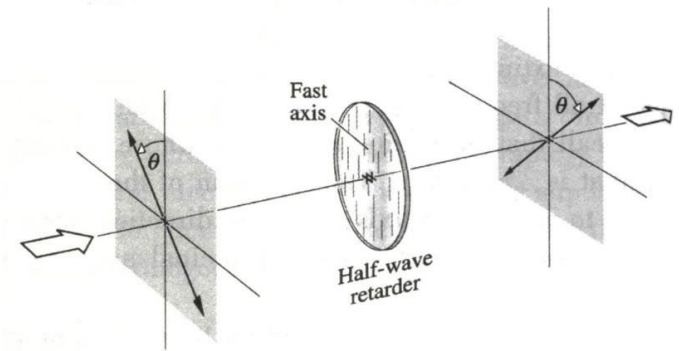
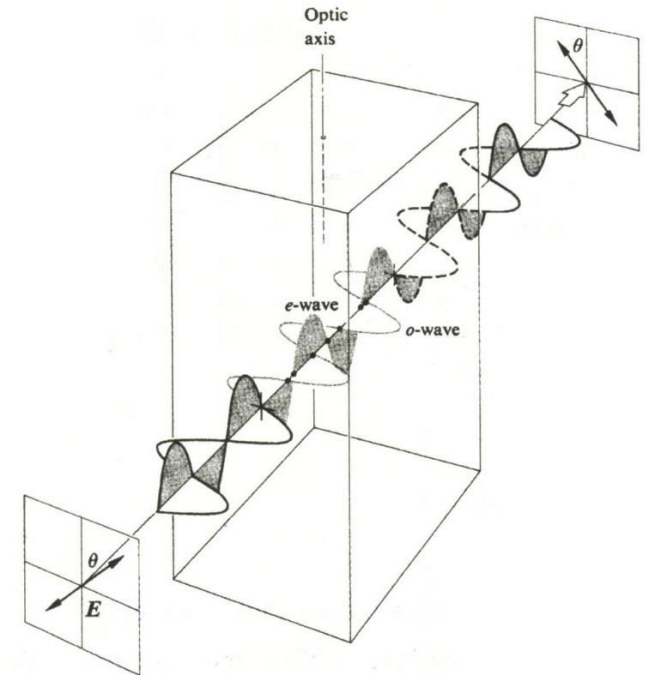
- ❑ As to elliptical light, it will have the same function.
- ❑ As to handedness of circular and elliptical light, it will invert.

The thickness of the materials must satisfy:

$$d(|n_o - n_e|) = (2m + 1) \lambda_0 / 2$$

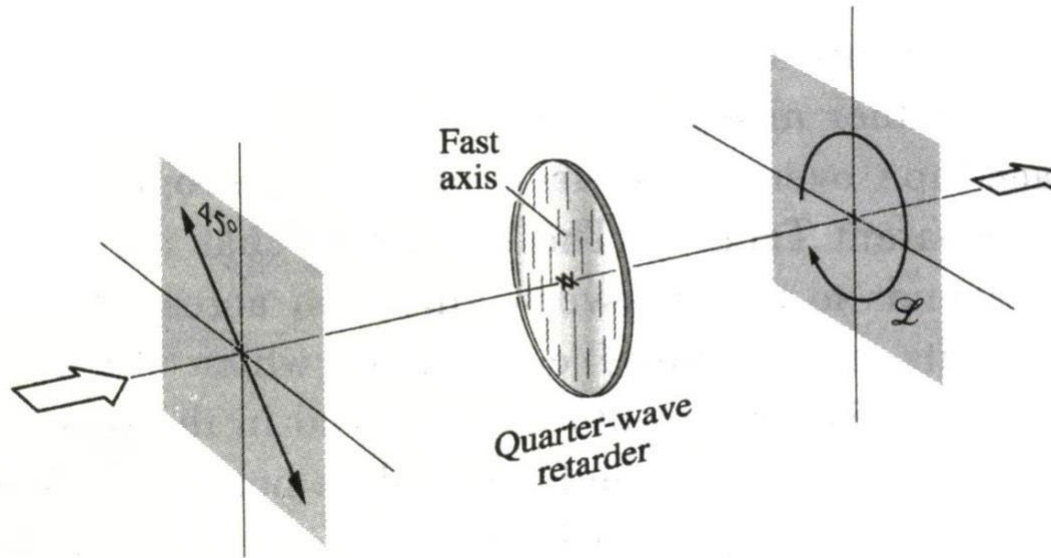
$$m = 0, 1, 2, \dots$$

Materials: Mica, crystalline quartz, single crystal magnesium fluoride (for IR), and cadmium sulfide (for IR)



The Quarter-Wave Plate

The Quarter-Wave Plate: $\Delta\varphi=\pi/2$ between the constituent orthogonal o- and e-components of a wave.



When linear light at 45° to either principle axis incident on a quarter-wave plate, its o- and e-components have equal amplitude, a 90° phase shift converts the wave into circular light.

The thickness of the birefringent materials must satisfy the expression:

$$d(|n_o - n_e|) = (2m + 1) \lambda_0 / 4$$

$$m = 0, 1, 2, \dots$$

Retarders (wave plates):

Zero-order retarder (零级波片): the minimum thickness necessary to produce the required phase difference.

For example: a quartz quarter-wave plate; Birefringence: 0.0092 at 550nm => thickness: 15 μ m.

Fragile and difficult to fabricate; but large angular field-of-view.

Multiple-order retarder (多级波片): a thickness that corresponds to a whole number of 2π phase shifts plus the desired $\Delta\varphi$.

Less expensive, but sensitive to wavelength, incident angle and temperature, and narrow field-of-view.

Compound zero-order (复合零级波片): combine two multiple-order retarders whose retardance difference yields the desired value of $\Delta\varphi$.

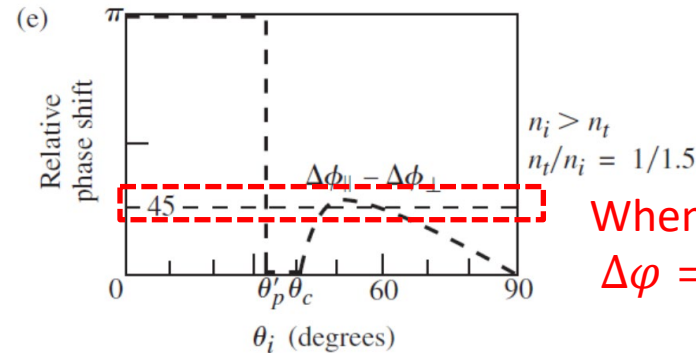
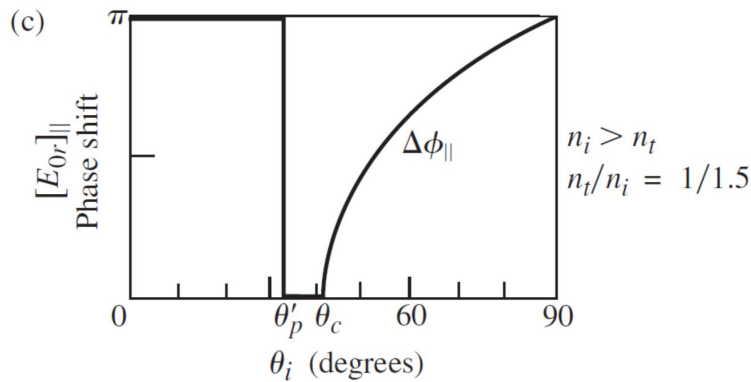
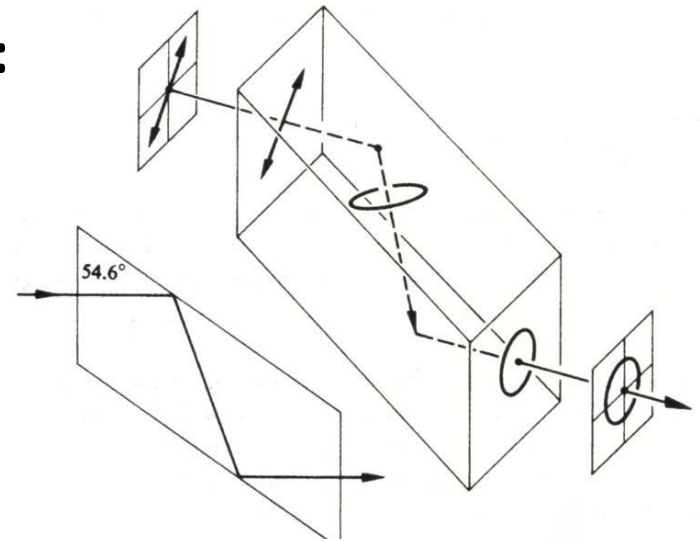
Tend to cancel the temperature variations

The Fresnel rhomb (菲涅尔斜方体) :

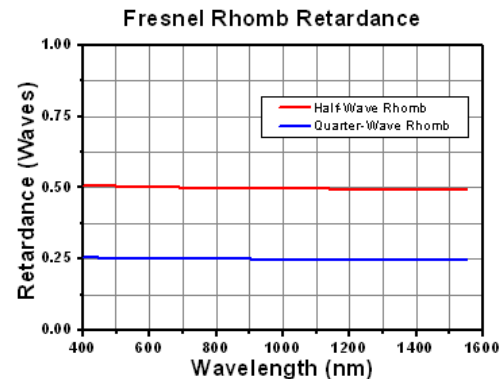
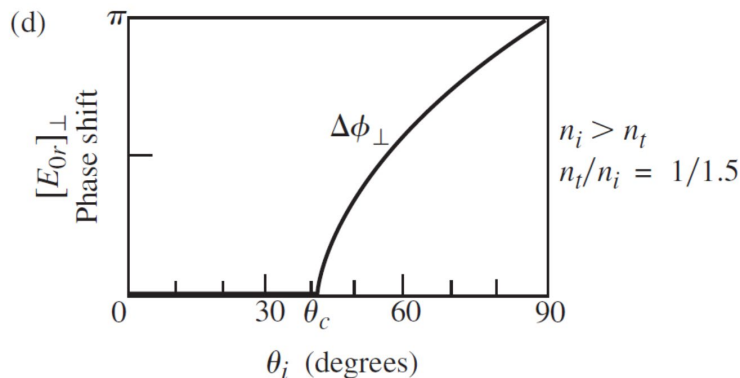
Principle: the total internal reflection introduced a relative phase difference between the two orthogonal field components

Note:

The retardance is almost independent of frequency over a larger range (全色的90°延迟器) .



When $\theta_i = 54.6^\circ$,
 $\Delta\phi = 45^\circ$

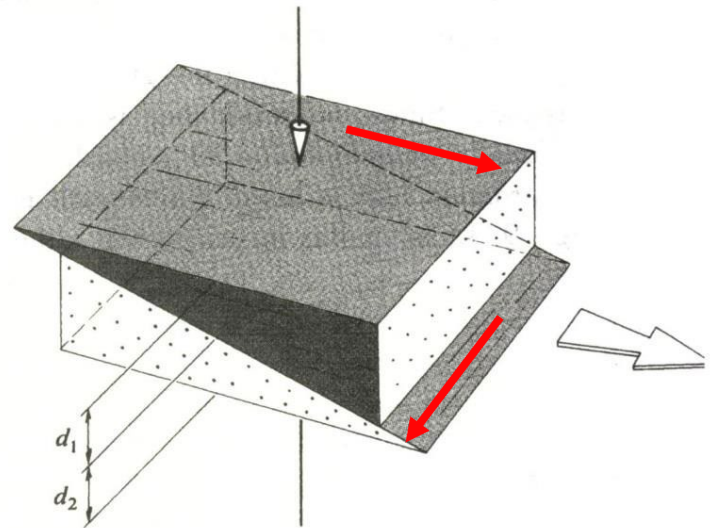


7.7.2 Compensators and variable retarders

A compensator is an optical device that is capable of impressing a controllable retardance on a wave.

Babinet compensator: the total phase difference

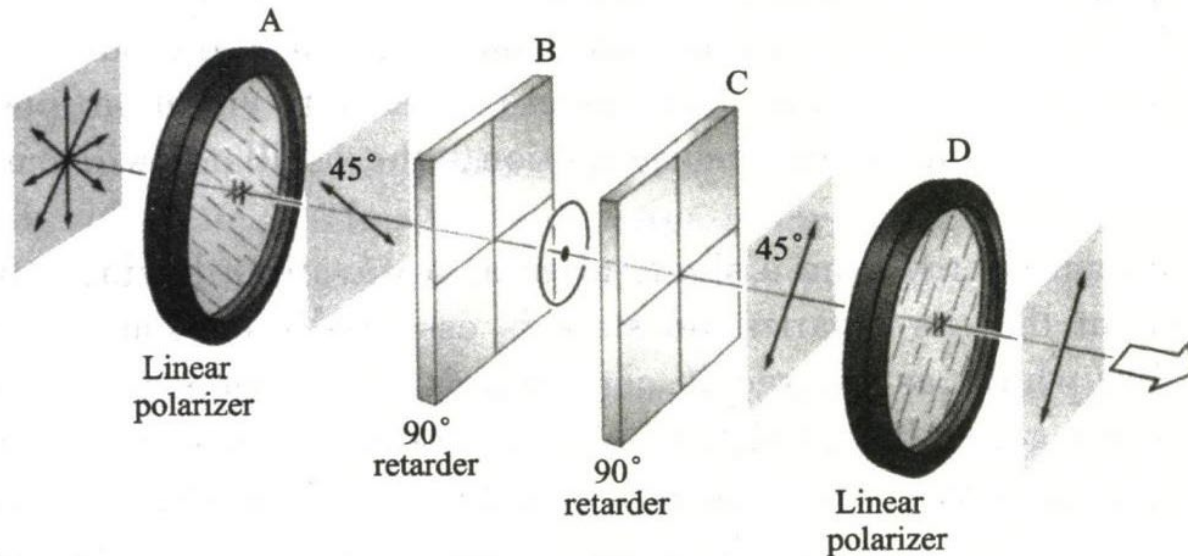
$$\Delta\varphi = \frac{2\pi}{\lambda_0} (d_1 - d_2)(|n_o - n_e|)$$



7.8 Circular Polarizers

Any series combination of an appropriately oriented linear polarizer and a 90° retarder will perform as a circular polarizer.

Example: It is used as an analyzer to determine the handedness of a wave that is already known to be circular.



- ☐ The Left-circular light entering a left-circular polarizer from the output side will be transmitted.
- ☐ The right-circular light will produce a P-state perpendicular to the transmission axis of D and so will be absorbed.

7.9 Polarization of polychromatic light

The radiant energy from a single atom:

- ❑ Wavetrain with a finite spatial extent Δl_c
- ❑ Assume its polarization state is constant within coherent time Δt_c

A typical source: a large collection of such radiating atoms, oscillating with different phases at some dominant frequency $\bar{\nu}$.

- ❑ In a very small region of the source and during very short time: the resultant wave has a polarization state.

Consider a wave whose bandwidth is very small in comparison with its mean frequency, a quasimonochromatic wave

$$\vec{E}_x(z, t) = \hat{i}E_{0x}\cos[\bar{k}z - \bar{\omega}t + \epsilon_x(t)]$$

$$\vec{E}_y(z, t) = \hat{j}E_{0y}\cos[\bar{k}z - \bar{\omega}t + \epsilon_y(t)]$$

- ❑ In a long time: the amplitude and phase will vary either independently or in some correlated fashion. So the wave is unpolarized.

7.9.2 Interference Colors

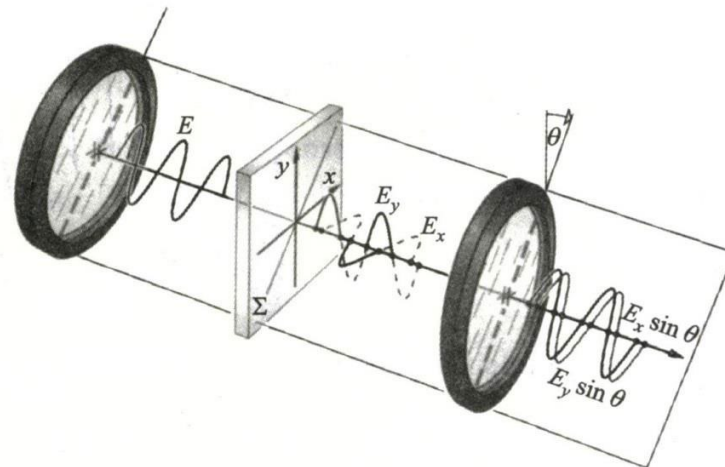
Interference colors arise from the wavelength dependence of the retardation.

$$\Delta\varphi = \frac{2\pi}{\lambda_0} d(|n_o - n_e|)$$

Example: the retardance arising some point P on retarder for blue light ($\lambda_0 = 435 \text{ nm}$) is 4π .

For incident yellow light ($\lambda_0 = 580 \text{ nm}$) is $\Delta\varphi = 3\pi$

If the blue light is parallel to the analyzer's transmission axis, the yellow light will along the extinction axis.



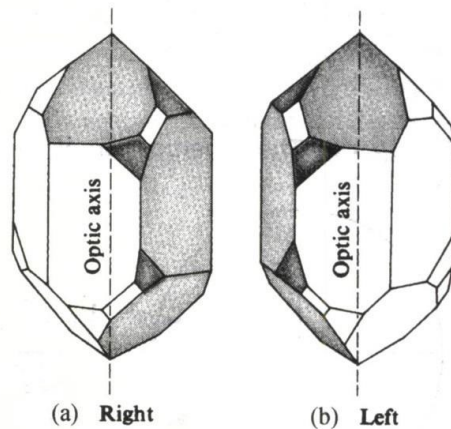
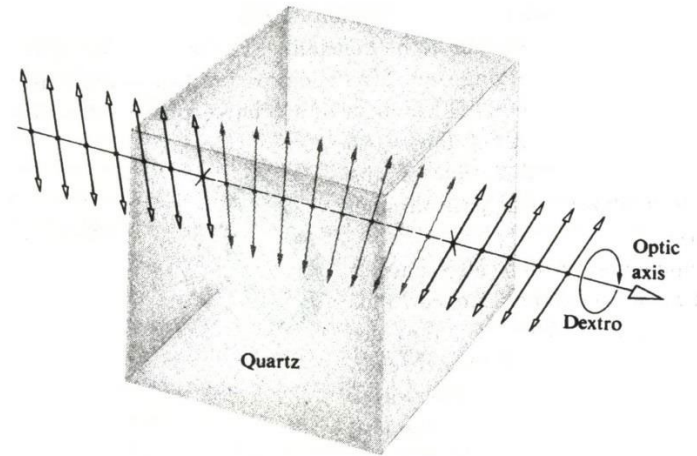
Interference colors from liquid crystals



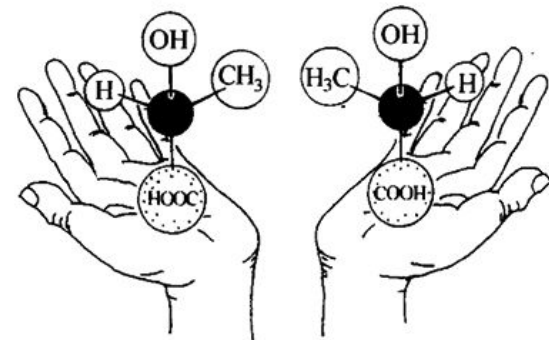
7.10 Optical Activity

Any material that causes the E-field of an incident linear plane wave to appear to rotate is said to be **optically active**.

- ❑ If the plane-of-vibration appears to have revolved clockwise: dextrorotatory or d-rotatory
- ❑ If it displaces counterclockwise: the materials is levorotatory or l-rotatory.



手性分子 chiral molecule 具有手性的分子，一般在一个C原子上按四面体方式连



接上4个互不相同的基团形成手性分子，如图示出左右两只手托出两个呈镜像关系的乳酸分子。一切具有螺旋结构的分子都是手性分子。[P]